



Design of an industry quality instrumentation amplifier ASIC using OS tools / silicon

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Quick introduction

- SLICE Semiconductor formed in 2024 to provide mixed signal IC design consulting & contract services in addition to low cost ASIC development (<https://www.slicesemiconductor.com/>)



- Since our inception we have served both SME's and MNC's globally
- SLICE Semiconductor is passionate about open source and truly believes in its importance to the semiconductor industry
- Our low cost ASIC development service is entirely based on an open source flow which has been shown to reduce ASIC development NRE costs by up to 10x

Motivation for this project

- To answer the question – is the open source infrastructure ready to produce industry quality products?
- We believe yes, it has now reached a point of majority to deliver low frequency, analog on top ASICs
- This project demonstrates this by developing an industry quality ASIC, verified to industry standards (PVT, MC, PEX) with similar performance to that of a related product from ADI
- None of this would have been possible without the support of nlnet foundation who funded the entire work



- Official project launch page: <https://nlnet.nl/project/INA-ASIC/>
- Due for tapeout on July 13th with IHP (SG13CMOS5L - €1.5k/mm² for 40 samples)
- Silicon returning on Dec. 16th (35 x QFN36 packages + 5 bare die)
- Full lab char commencing Jan. '27



ASIC introduction

- ASIC is a programmable gain instrumentation amplifier (INA) (Project name = Sunrise)
- Design specifications are shown below

Parameter	Unit	Value	Comment
vdda_hv	V	3.3	Analog supply
vdda_lv	V	1.5	Digital supply
vssa	V	0	Ground
idda_hv	mA	1	Analog supply current
Power	mW	3.465	Analog power consumption
Gain error	%	0.5	INA Accuracy within $\pm 50\text{mV}$ of the supply
CMRR	dB	80	Common mode rejection ratio
Noise density	nV/rHz	50	Measured at 1kHz
Integrated noise	uVrms	200	Set by temp sense requirements
Bandwidth	kHz	50	
LG(0)min	dB	70	For all -ve feedback loops
PM min	deg	55	10deg > post layout spec to give margin
GM min	dB	13	3dB > post layout spec to give margin
Noise density	nV/rHz	30	For all -ve feedback loops
CMRR(0) min	dB	120	Applicable to pamps in all -ve feedback loops
idda_hv	uA	300	For all -ve feedback loops

- The “big 3” specs are highlighted
- These drove the design and make the INA performance similar to ADI’s AD8223A [1]
- INA has 4 gain settings (1 / 5 / 11 /21) programmable via I2C / SPI interfaces
- ASIC also contains an on-chip temp sensor which has been reported to measure down below -200C [2]
- ASIC also contains a 6b tunable 10MHz ring oscillator covering a $\pm 20\%$ range down to 1% accuracy
- Finally, physical design has been implemented to make the ASIC robust to high radiation environments (e.g. LEO)

Tools, project repo and supply chain

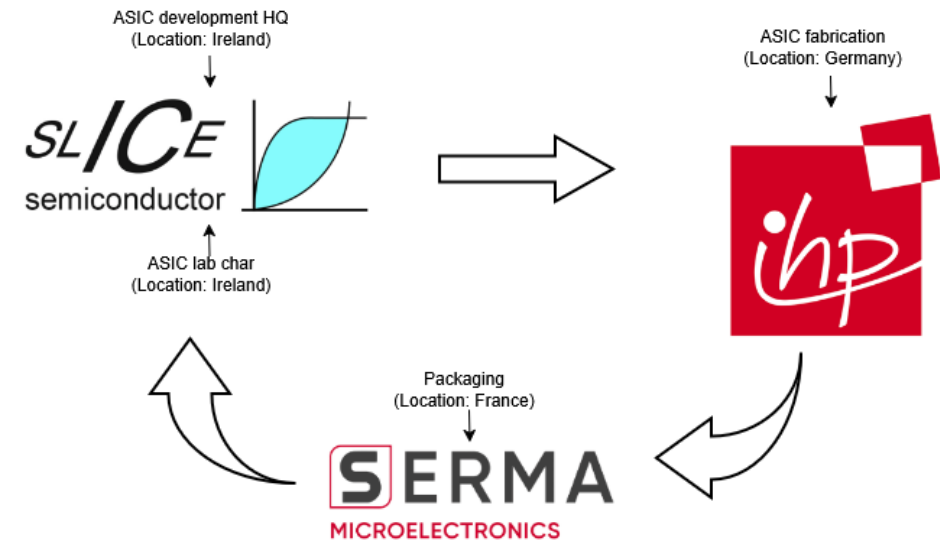
- Tool chain used to develop this ASIC



#	Tool	Function
1	ngspice	SPICE simulator
2	xschem	schematic entry
3	xschem	waveform viewer
4	klayout	physical design / DRC
5	magic	pycell generation / DRC / PEX
6	netgen	LVS
7	openVAF	verilogA
8	Octave	post processing
9	Icarus Verilog	Digital simulation
10	Libre Lane	Physical implementation
11	Surfer	waveform viewer

Analog
Digital

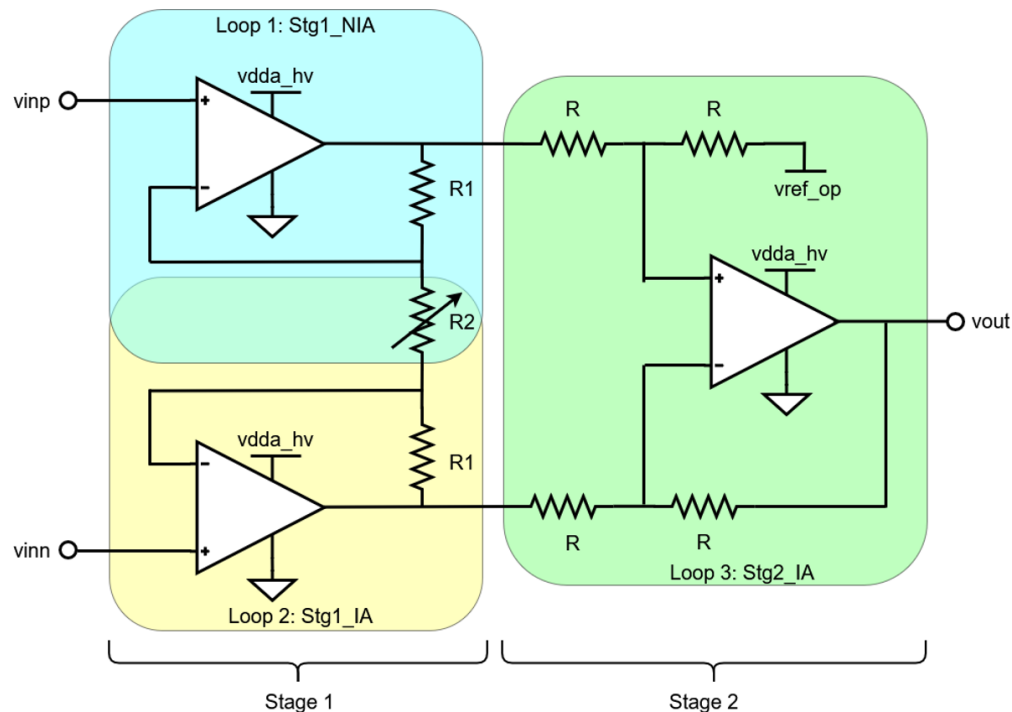
- Supply chain used to develop this ASIC



- Project repo containing schematics / layout / design review reports found at the below link:
- https://github.com/SLICESemiconductor/Space_grade_Instrumentation_Amplifier_ASIC/tree/main

Brief intro to INA theory

- INA's can be discrete or continuous time based
- INA for this ASIC is continuous time, based on the conventional 3-opamp architecture shown below



- INA thus composed of 2 stages, comprising 3 loops, which utilize 2 distinct –ve feedback loops
- Gain can be shown to be [3]:

$$A_{v_{INA}} = \left(1 + \frac{2R_1}{R_2}\right)$$

- Gain error can be shown to be [3]:

$$GE \approx \frac{1}{LG_{stg1}} + \frac{1}{LG_{stg2}} + \frac{r_{on}}{R_2}$$

- CMRR can be shown to be [3]:

$$CMRR = 20 \log_{10} \left(\frac{e}{A_v}\right)$$

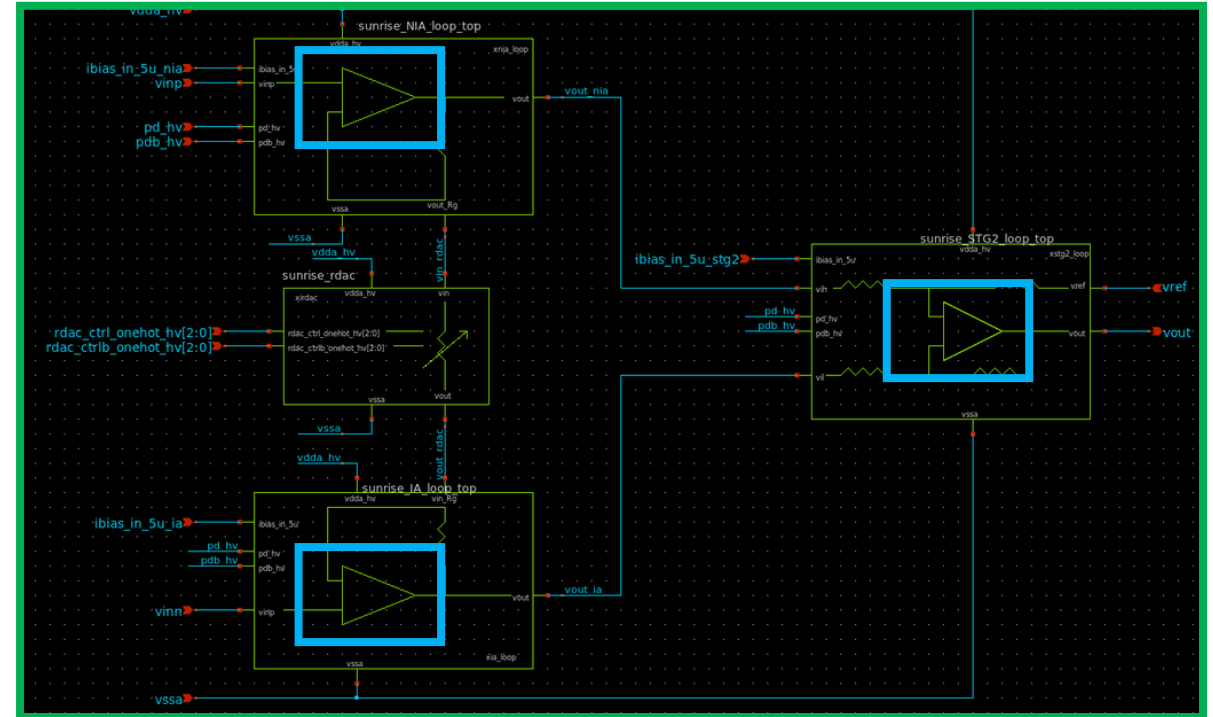
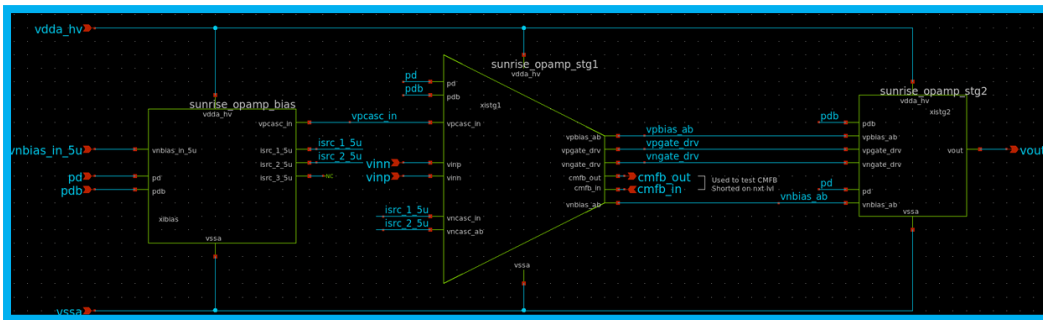
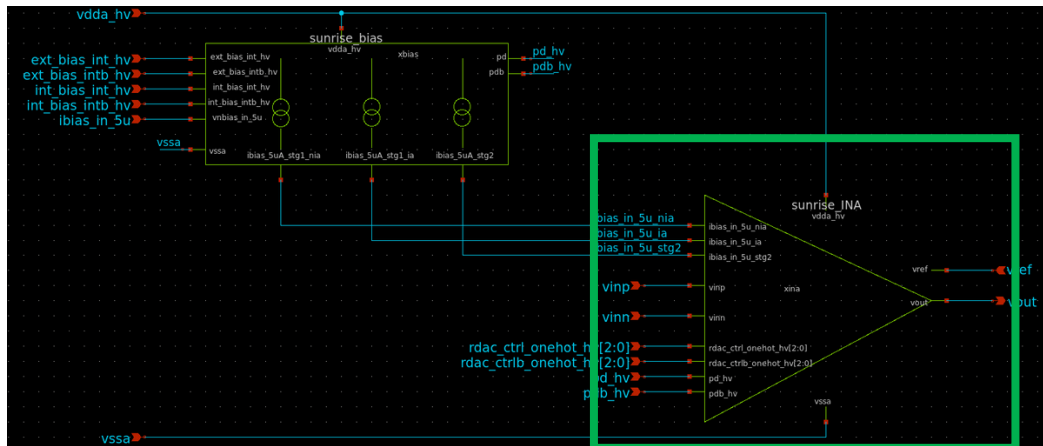
- Noise can be shown to be [3]:

$$v_{n_{opr}}^2 = 2.4kTR_{eq1}NG_1^2 + 2.4kTR_{eq1}SG_3^2 + 2.4kTR_{eq5}NG_5^2$$

$$v_{n_{ipr}} = \sqrt{v_{n_{opr}}^2 / A_{v_{ina}}}$$

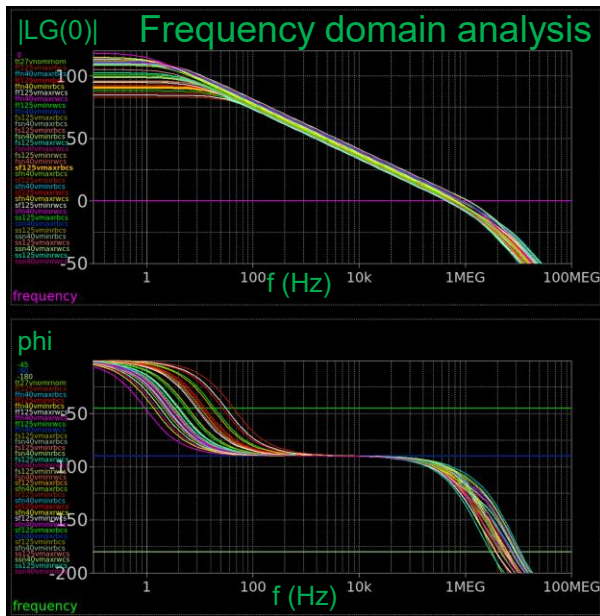
INA implementation (schematics)

- INA schematic layered into multiple hierarchies to aid design / verification / post layout debug
- Therefore, to give a feel for the schematic entries in xschem, below shows a representative portion



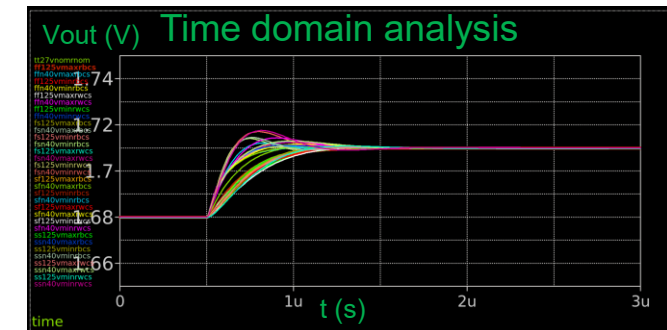
INA results – Stability

- As per industry standard, stability was verified in frequency / time domains across PVT
- Below shows such analysis for one of the INA loops (stage 1 NIA loop)



Frequency stb PVT data

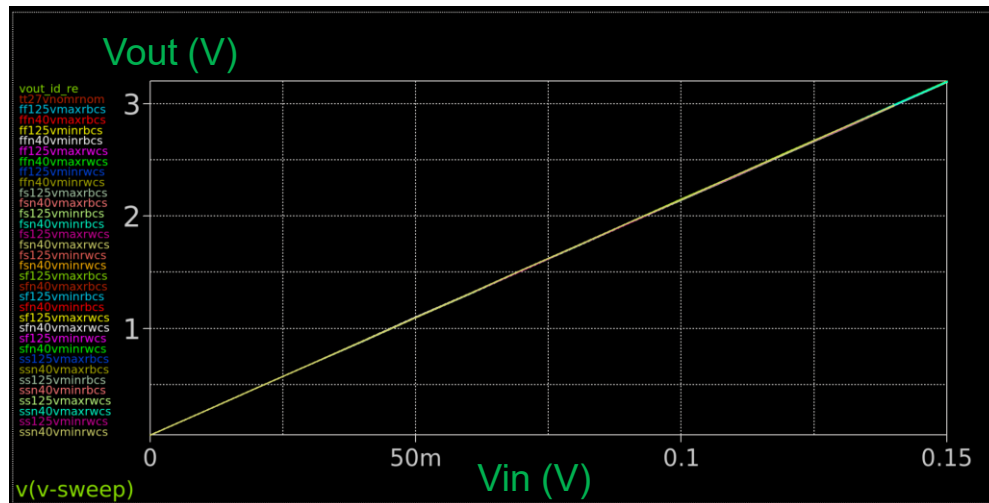
cor_id	T_degC	vdda_lv	vdda_lv	LG0_dB	BW_kHz	GBW_MHz	PM_deg	GM_dB
0	27	3.3	1.5	108.079	0.0025	0.6281	69.9948	-27.2043
25	125	3.465	1.575	82.761	0.043	0.5741	75.5368	-25.6342
9	-40	3.465	1.575	112.157	0.0018	0.7984	76.2561	-27.4042
17	125	3.135	1.425	86.2067	0.0269	0.5319	73.2566	-23.5933
1	-40	3.135	1.425	108.08	0.0033	0.833	70.7915	-26.3988
29	125	3.465	1.575	84.3998	0.0308	0.4872	73.1956	-26.9813
13	-40	3.465	1.575	114.627	0.0009	0.6523	71.2221	-29.4131
21	125	3.135	1.425	87.9461	0.019	0.4528	71.0727	-25.548
5	-40	3.135	1.425	109.534	0.0023	0.6638	66.4033	-28.7633
28	125	3.465	1.575	95.438	0.0104	0.5973	72.8009	-24.2648
12	-40	3.465	1.575	110.869	0.0024	0.8407	72.236	-26.9511
20	125	3.135	1.425	95.8991	0.0094	0.5665	70.1687	-20.6347
4	-40	3.135	1.425	108.964	0.0041	1.1542	61.6962	-19.0332
32	125	3.465	1.575	98.8156	0.0061	0.5047	70.3633	-25.9525
16	-40	3.465	1.575	112.689	0.0015	0.6808	67.3838	-29.1083
24	125	3.135	1.425	98.4656	0.006	0.4777	67.8983	-22.7755
8	-40	3.135	1.425	110.427	0.0029	0.9	57.7098	-21.8084
27	125	3.465	1.575	90.5514	0.0171	0.5595	74.1042	-24.9202
11	-40	3.465	1.575	111.031	0.0021	0.7801	74.4452	-27.4059
19	125	3.135	1.425	92.5245	0.0126	0.5191	71.7569	-21.5394
3	-40	3.135	1.425	108.875	0.0033	0.8995	66.7357	-24.3521
31	125	3.465	1.575	92.676	0.0116	0.4758	71.8134	-26.5247
15	-40	3.465	1.575	112.755	0.0014	0.6371	69.5612	-29.4786
23	125	3.135	1.425	94.5671	0.0087	0.4425	69.5964	-23.7238
7	-40	3.135	1.425	110.217	0.0023	0.7036	62.6754	-27.3263
26	125	3.465	1.575	100.807	0.0054	0.5852	71.2829	-22.7246
10	-40	3.465	1.575	110.403	0.0026	0.8397	69.534	-26.683
18	125	3.135	1.425	99.5547	0.006	0.5597	68.4674	-18.52
2	-40	3.135	1.425	109.447	0.0039	1.1245	59.4422	-17.7019
30	125	3.465	1.575	104.384	0.003	0.4947	68.9141	-24.7497
14	-40	3.465	1.575	111.843	0.0018	0.6789	64.8964	-28.8914
22	125	3.135	1.425	102.076	0.0038	0.4719	66.2329	-20.3725
6	-40	3.135	1.425	110.87	0.0028	0.8845	55.4467	-20.143
MIN	MIN	MIN	MIN	MIN	MIN	MIN	MIN	MIN
0	-40	3.135	1.425	82.761	0.0009	0.4425	55.4467	-29.4786
MAX	MAX	MAX	MAX	MAX	MAX	MAX	MAX	MAX
32	125	3.465	1.575	114.627	0.043	1.1542	76.2561	-17.7019
absvar	absvar	absvar	absvar	absvar	absvar	absvar	absvar	absvar
32	165	0.33	0.15	31.866	0.0421	0.7117	20.8094	11.7767
relvar_pmpcr	relvar_pmpcr	relvar_pmpcr	relvar_pmpcr	relvar_pmpcr	relvar_pmpcr	relvar_pmpcr	relvar_pmpcr	relvar_pmpcr
100	194.1176	5	5	16.1438	95.7292	44.5715	15.8003	-24.9609



- As shown, PM_min = 55deg and GM_min = -17.7dB
- No ringing observed in the time domain response
- The above was run at SCH level. Only TT / WC were run at PEX (industry standard)
- After 2 layout iterations (of the compensation cap) pre and post layout results matched to within 1deg

INA results – Gain error

- Gain error was verified across PVT
- Below shows such analysis for a gain setting of 11
- Output voltage is swept from 50mV → vdda -50mV



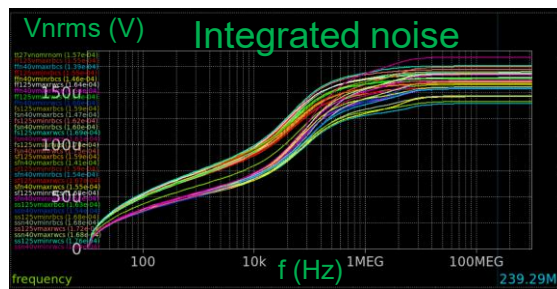
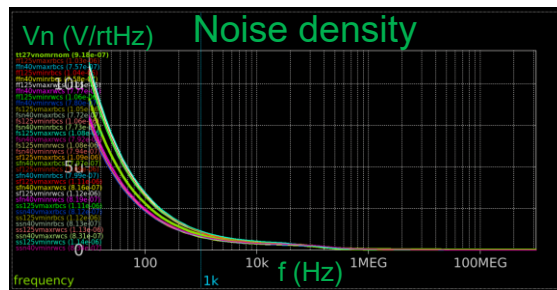
- As shown, worst case gain error = 0.4%
- Therefore, gain error 0.5% spec verified to be met across PVT
- Running PEX on TT / WC corners showed no significant degradation (0.5% spec still met)

Gain error PVT data

cor_id	T_degC	vdda_hv	vdda_lv	ge_per
0	27	3.3	1.5	-0.1402
25	125	3.465	1.575	-0.3357
9	-40	3.465	1.575	-0.0992
17	125	3.135	1.425	-0.3409
1	-40	3.135	1.425	-0.1119
29	125	3.465	1.575	-0.2417
13	-40	3.465	1.575	-0.0615
21	125	3.135	1.425	-0.2326
5	-40	3.135	1.425	-0.0755
28	125	3.465	1.575	-0.2949
12	-40	3.465	1.575	-0.1109
20	125	3.135	1.425	-0.3417
4	-40	3.135	1.425	-0.1349
32	125	3.465	1.575	-0.187
16	-40	3.465	1.575	-0.0686
24	125	3.135	1.425	-0.2151
8	-40	3.135	1.425	-0.0849
27	125	3.465	1.575	-0.312
11	-40	3.465	1.575	-0.1124
19	125	3.135	1.425	-0.3511
3	-40	3.135	1.425	-0.1378
31	125	3.465	1.575	-0.2057
15	-40	3.465	1.575	-0.0695
23	125	3.135	1.425	-0.2255
7	-40	3.135	1.425	-0.0867
26	125	3.465	1.575	-0.3238
10	-40	3.465	1.575	-0.1279
18	125	3.135	1.425	-0.3936
2	-40	3.135	1.425	-0.1597
30	125	3.465	1.575	-0.2005
14	-40	3.465	1.575	-0.0788
22	125	3.135	1.425	-0.244
6	-40	3.135	1.425	-0.0998
MIN	MIN	MIN	MIN	MIN
0	-40	3.135	1.425	-0.3936
MAX	MAX	MAX	MAX	MAX
32	125	3.465	1.575	-0.0615
absvar	absvar	absvar	absvar	absvar
32	165	0.33	0.15	0.3321
relvar_pmpcr	relvar_pmpcr	relvar_pmpcr	relvar_pmpcr	relvar_pmpcr
100	194.1176	5	5	-72.988

INA results – Noise

- Noise (density + integrated) was verified across PVT
- Below shows such analysis for a gain setting of 21



Noise density PVT data

cor_id	T degC	vdda_lv	vdda_hv	vn_ipr_1k_nVrtHz
0	27	3.3	1.5	43.695
25	125	3.465	1.575	49.0791
9	-40	3.465	1.575	36.0342
17	125	3.135	1.425	49.4034
1	-40	3.135	1.425	36.1032
29	125	3.465	1.575	50.2061
13	-40	3.465	1.575	37.0058
21	125	3.135	1.425	50.608
5	-40	3.135	1.425	37.1253
28	125	3.465	1.575	50.1803
12	-40	3.465	1.575	36.7422
20	125	3.135	1.425	50.4542
4	-40	3.135	1.425	36.8116
32	125	3.465	1.575	51.3003
16	-40	3.465	1.575	37.6939
24	125	3.135	1.425	51.6575
8	-40	3.135	1.425	37.8163
27	125	3.465	1.575	51.8076
11	-40	3.465	1.575	37.9652
19	125	3.135	1.425	52.096
3	-40	3.135	1.425	38.0288
31	125	3.465	1.575	52.8942
15	-40	3.465	1.575	38.8781
23	125	3.135	1.425	53.2707
7	-40	3.135	1.425	38.9972
26	125	3.465	1.575	52.8332
10	-40	3.465	1.575	38.6744
18	125	3.135	1.425	53.0973
2	-40	3.135	1.425	38.7368
30	125	3.465	1.575	53.9148
14	-40	3.465	1.575	39.57
22	125	3.135	1.425	54.2747
6	-40	3.135	1.425	39.6908
MIN MIN MIN MIN MIN				
0	-40	3.135	1.425	36.0342
MAX MAX MAX MAX MAX				
32	125	3.465	1.575	54.2747
absvar	absvar	absvar	absvar	absvar
32	165	0.33	0.15	18.2405
relvar_pmpcr	relvar_pmpcr	relvar_pmpcr	relvar_pmpcr	relvar_pmpcr
100	194.1176	5	5	20.1979

Integrated noise PVT data

cor_id	T degC	vdda_lv	vdda_hv	vn_int_uV
0	27	3.3	1.5	156.637
25	125	3.465	1.575	153.899
9	-40	3.465	1.575	138.726
17	125	3.135	1.425	154.023
1	-40	3.135	1.425	145.314
29	125	3.465	1.575	163.253
13	-40	3.465	1.575	153.378
21	125	3.135	1.425	162.589
5	-40	3.135	1.425	159.107
28	125	3.465	1.575	158.381
12	-40	3.465	1.575	146.087
20	125	3.135	1.425	160.725
4	-40	3.135	1.425	159.362
32	125	3.465	1.575	167.745
16	-40	3.465	1.575	160.368
24	125	3.135	1.425	169.049
8	-40	3.135	1.425	173.698
27	125	3.465	1.575	157.673
11	-40	3.465	1.575	140.696
19	125	3.135	1.425	158.622
3	-40	3.135	1.425	152.624
31	125	3.465	1.575	166.447
15	-40	3.465	1.575	154.454
23	125	3.135	1.425	166.625
7	-40	3.135	1.425	165.999
26	125	3.465	1.575	162.411
10	-40	3.465	1.575	153.594
18	125	3.135	1.425	166.89
2	-40	3.135	1.425	167.359
30	125	3.465	1.575	171.216
14	-40	3.465	1.575	167.122
22	125	3.135	1.425	174.592
6	-40	3.135	1.425	182.932
MIN MIN MIN MIN MIN				
0	-40	3.135	1.425	138.726
MAX MAX MAX MAX MAX				
32	125	3.465	1.575	182.932
absvar	absvar	absvar	absvar	absvar
32	165	0.33	0.15	44.206
relvar_pmpcr	relvar_pmpcr	relvar_pmpcr	relvar_pmpcr	relvar_pmpcr
100	194.1176	5	5	13.7432

- Marginal spec violation of 54nV/rtHz not addressed since it occurred at 125C. Since AD8223A noise spec is quoted at 25C, re-running the skew corner at 25C showed noise < 50nV => extra area not justified
- Integrated noise well below the 200uVrms spec
- Running PEX on TT / WC corners showed no significant degradation (pretty typical with noise)

INA results – CMRR

- Opamps show CMRR's < 120dB
- Overall INA CMRR limited by stage 2 resistor mismatch (i.e. random and not systematic)
- Below shows CMRR results across MC spread (30 runs) under various conditions

$A_v = 5$



T (degC)	External bias	Internal bias
-40	82.8	81.1
27	81.5	82.3
125	81.4	81.2

$A_v = 11$



T (degC)	External bias	Internal bias
-40	83.1	83.2
27	84.9	83.7
125	83.1	85.2

$A_v = 21$

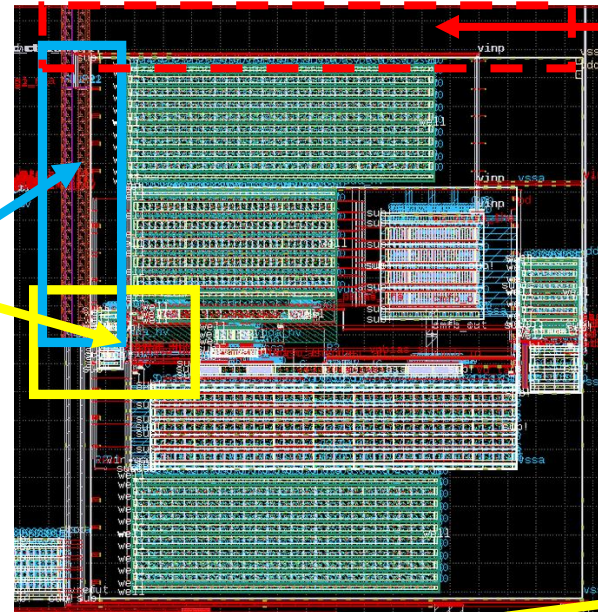
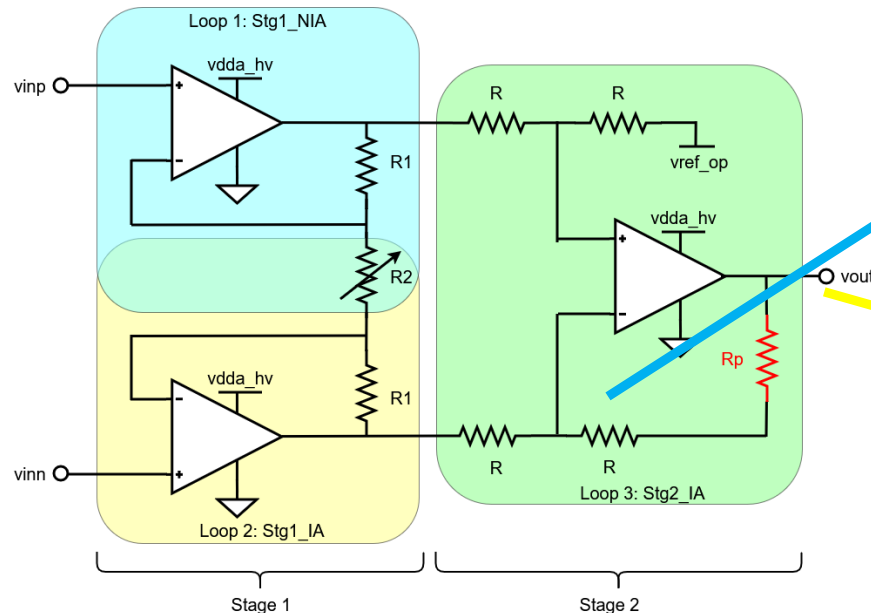


T (degC)	External bias	Internal bias
-40	84.5	86.3
27	85.5	85.7
125	84.3	84.8

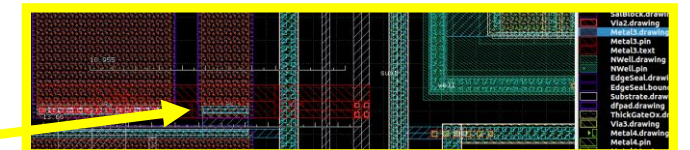
- Note the use of the External / Internal bias terminology. This is because the INA can be biased using an internal reference or externally from the bench
- As shown, the 80dB CMRR spec is met in all scenarios
- Above tested at SCH
- Re-running PEX showed a different story ... see next slide!

INA results – CMRR

- Because of the opamps 120dB performance, systematic CMRR of the entire INA = 114dB (TT/SCH)
- Initial PEX run showed a systematic CMRR of 57dB!!!
- Root cause of this 57dB loss found to be due to incorrect placement of a resistor in the output stage



- Original resistor location
- Resistor i/p too far from opamp o/p to create a large parasitic (R_p)
- Resulting offset had a common mode dependency to kill overall CMRR



- Fix places resistor at the opamp output (10um M3 route) to remove the offset and regain our high CMRR
- This debug showed the value / accuracy and debug capability of the open source PEX flow

INA results - Other

- Many other results exist where the interested reader is referred to the full design report at [4]
- Below summarises the results across PVT for 5 / 11 / 21 gain settings with internal / external biasing

Parameter	Gain (x)							Spec met (Y/N)	AD8223A	Matches AD8223A
	5	5	11	11	21	21	Spec			
Biasing	External	Internal	External	Internal	External	Internal	Both	--		Y
Gain error (%)	0.33	0.33	0.37	0.37	0.39	0.39	0.5	Y	0.3	Y
$v_{n_1kHz_max}$ (nV/√Hz) ³	62 (50)	62 (50)	56 (46)	56 (46)	54 (44)	54 (44)	50	N	50 ^{1,2}	Y
v_{n_rms} (μV)	153	163	167	156	188	192	200	Y	--	Y
CMRR (dB)	81.4	81.1	83.1	83.2	84.3	84.8	80	Y	80	Y
BW (kHz)	176	177	94	87	55	51	50	Y	125 ¹ / 70 ¹	Y
i_{dda} (mA)	0.802	0.954	0.806	0.957	0.814	0.966	1	Y	0.5	N
1. Performance only provided for TT/25°C										
2. Quoted for gain = 5										
3. TT/27°C numbers shown in brackets										

- In addition, the above compares performance to AD8223A showing it to match all the relevant specs (gain error, noise, CMRR, BW)
- However it did require 2x more power (1mA instead of 0.5mA)
- This is due to the fact that the same opamp was used in each stage
- Given more time we could have optimised different opamps for each stage hence dramatically reducing total power

Temperature sensor theory

- On-chip temp sensors typically based on the physical property of thermal voltage

$$v_t = \frac{kT}{q} \Rightarrow \frac{d(v_t)}{dT} = \frac{k}{q} = 86\mu V/^{\circ}C$$

- Can make practical use of this property by measuring the difference in base-emitter voltages of 2 BJTs (biased by a current ratio = M) to give:

$$\Delta v_{be} = n \cdot v_t \cdot \ln(M) = n \cdot \frac{kT}{q} \cdot \ln(M) \Rightarrow \frac{d(\Delta v_{be})}{dT} = n \cdot \ln(M) \cdot \frac{k}{q} = 198\mu V/^{\circ}C \quad (n = 1, m = 10)$$

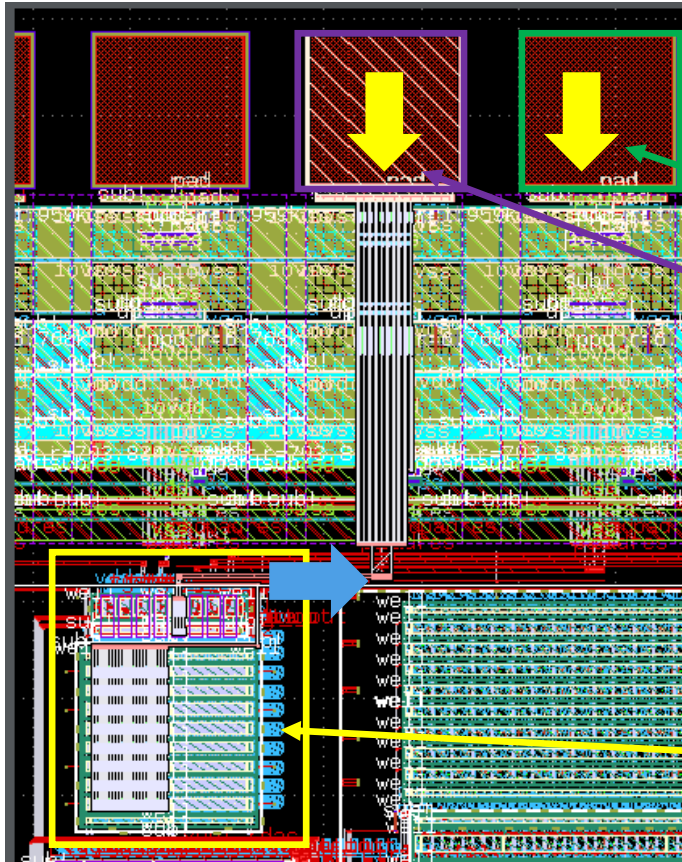
- Instead of using BJTs, we use MOSFETs biased in weak inversion to measure temperature using Δv_{gs}

$$\Delta v_{gs} = n \cdot v_t \cdot \ln(M) = n \cdot \frac{kT}{q} \cdot \ln(M) \Rightarrow \frac{d(\Delta v_{gs})}{dT} = n \cdot \ln(M) \cdot \frac{k}{q} = 198\mu V/^{\circ}C \quad (n = 1, m = 10)$$

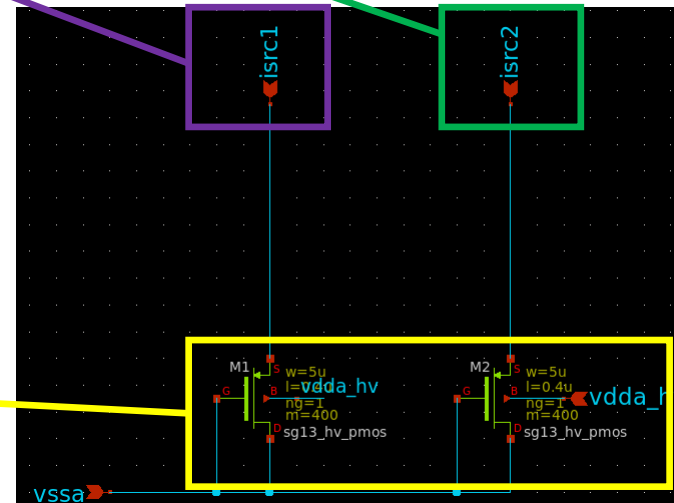
- Advantage of using Δv_{gs} is it remains linear to much lower temperatures (below -200C)
- Resolution of $\approx 200\mu V/C$ is too small to measure in the presence of circuit non-idealities
- The INA thus acts as the analog front end (AFE) to gain Δv_{gs} up by 21x and give a resolution of 4.2mV/C
- A resolution of 4.2mV/C is much easier to measure in the presence of circuit non-idealities

Temperature sensor implementation

- Below shows the implementation of the temp sensors

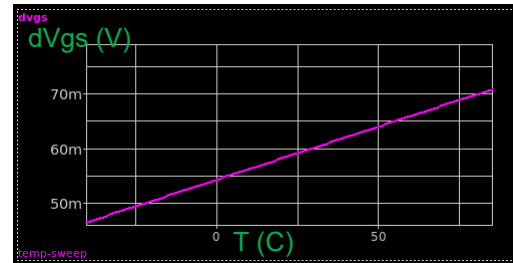
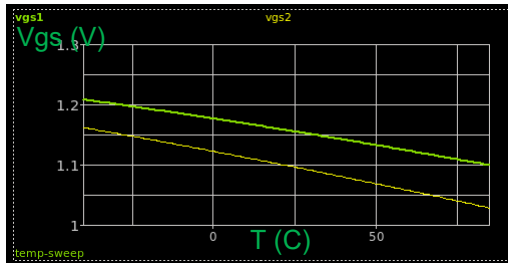


- Current sourced into the diodes through the IO's (yellow arrows)
- Δv_{gs} then sensed by the INA at the blue arrow, which it gains up by 21x



Temperature sensor results

- Below shows the v_{gs} of both diodes and their corresponding Δv_{gs} (which can be seen to be very linear)



- Linear co-efficients to characterize Δv_{gs} and $\Delta v_{gs} \times 21$ are:

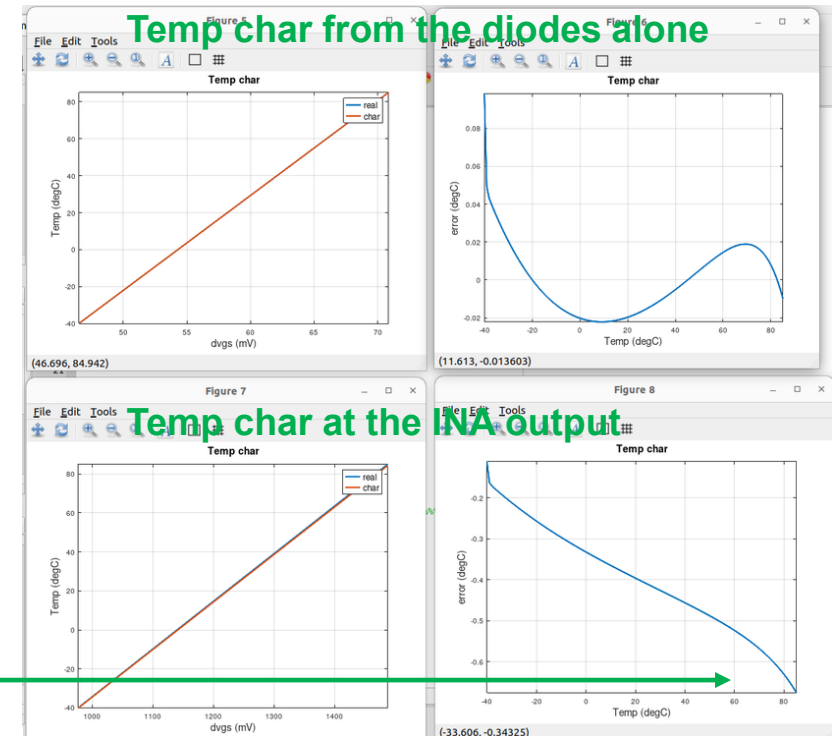
$m_char: 5131.61$
 $os_char: -278.609$
 $m_char\ opr: 244.36$

- Post processing shows the temp error from the diodes alone to be $\sim 0.1C$
- As expected, INA gain error grows this $0.6C$

$$\Delta v_{gs} = 1. \ln(10) \cdot ((1.38 \times 10^{-23} \cdot (273 + 85)) / 1.6 \times 10^{-19}) = 71.1mV \Rightarrow \Delta v_{outINAid} = 21 \cdot \Delta v_{gs} = 1.494V$$

$$GE \text{ (shown corner)} = 0.17\% \Rightarrow \Delta v_{outINAre} = 1.494 \times (0.17/100) = 2.54mV$$

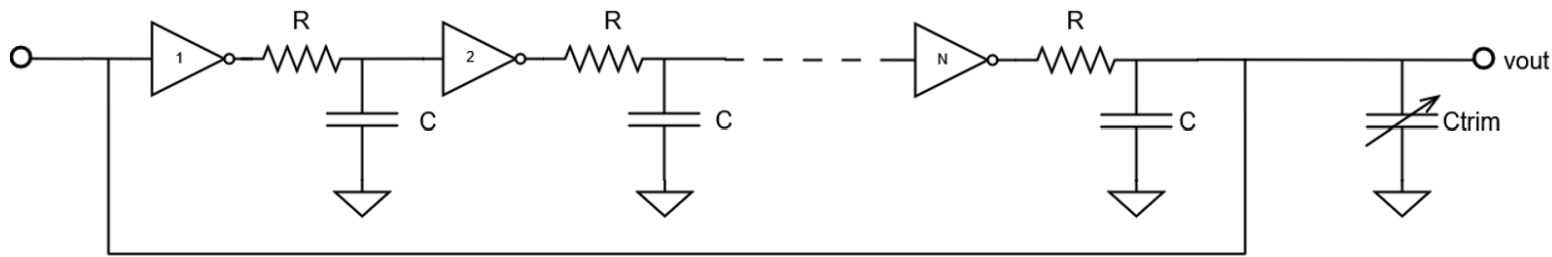
$$\text{At } d(v_{outINA})/dT = 4.2mV/^{\circ}C \Rightarrow T_e = (2.54/4.2) = 0.6^{\circ}C$$



- $0.6C$ temp error correlates with theory
- Achieves $\pm 1C$ target accuracy

Ring oscillator theory

- Shown is a standard 'N' stage low frequency RO
- RC components required to reduce frequency to MHz range
- Brute force (non-linear) trim achieved with a CDAC on the output

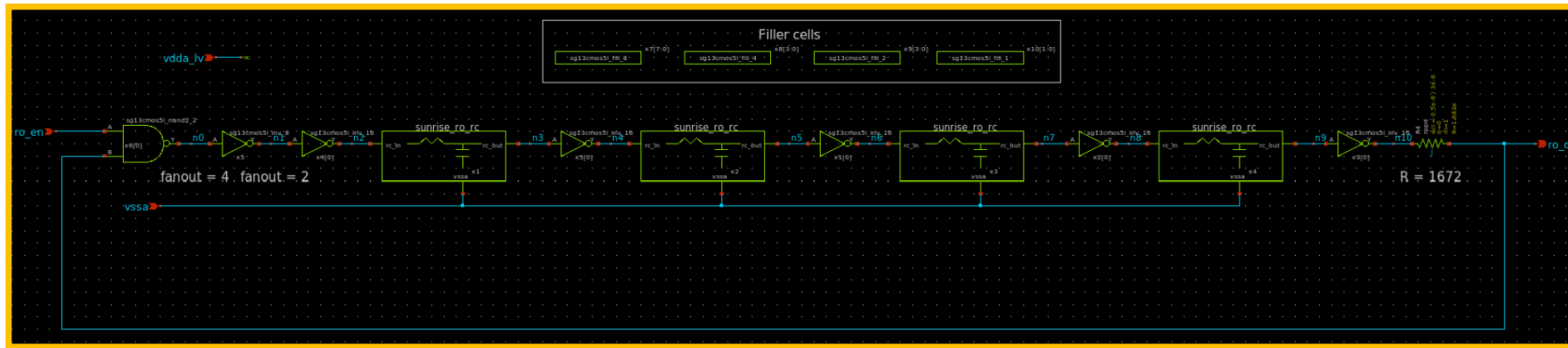
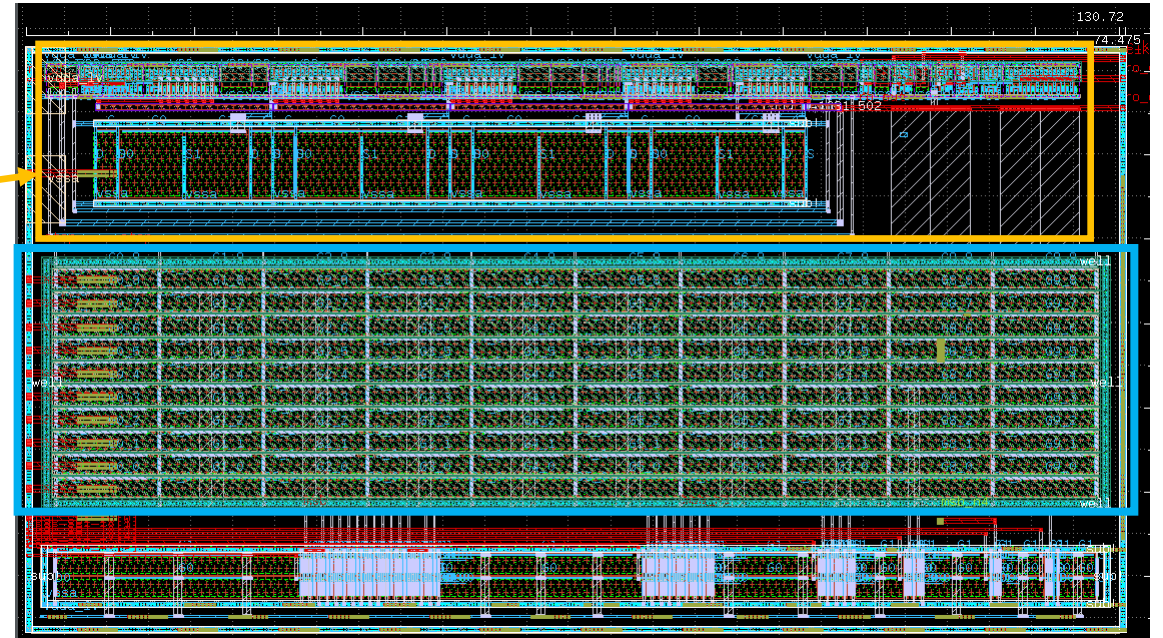
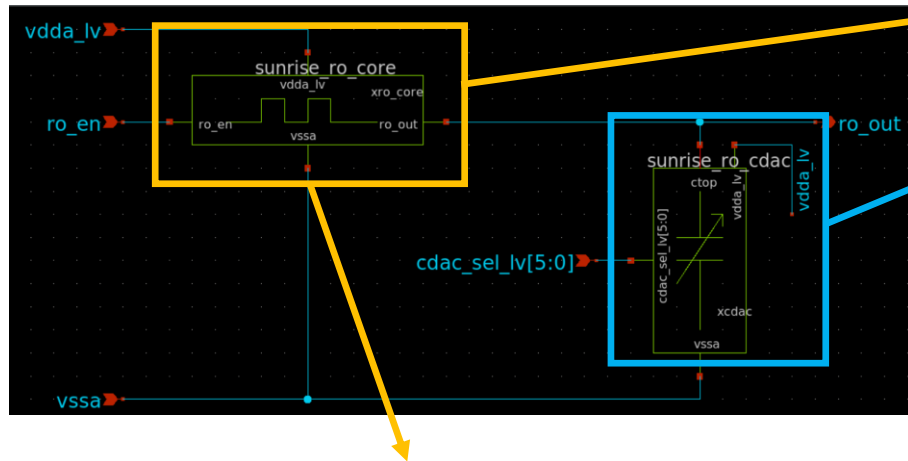


$$f_{osc} = \frac{1}{C_{trim} + (2 * 0.693 * RC * N)}$$

- Simple design (took 1 day to complete)
- Excellent for pipe cleaning the flow since it contains the full array of devices available
 - Std cells (inverters)
 - Passive components (RC elements)
 - Active components (CDAC)
- Also straight forward to measure and characterise post silicon in addition to be a quick visual demonstration of the ASICs d2a capabilities

Ring oscillator implementation

- Shown is a representative portion of the oscillator schematics and the top level layout

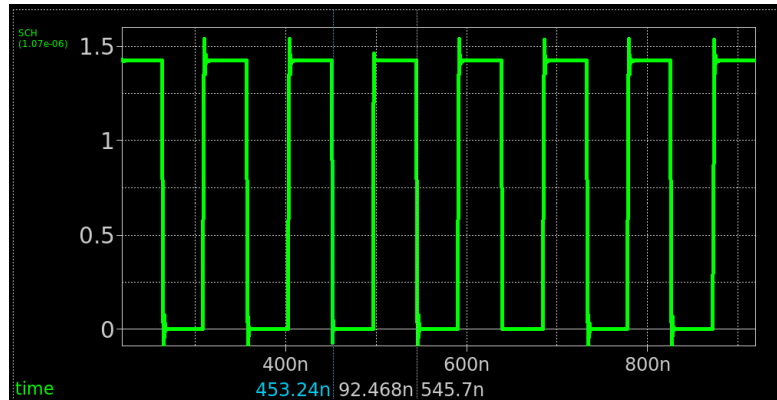


- As shown, layout occupies a footprint of ~ 130 x 75um

Ring oscillator results

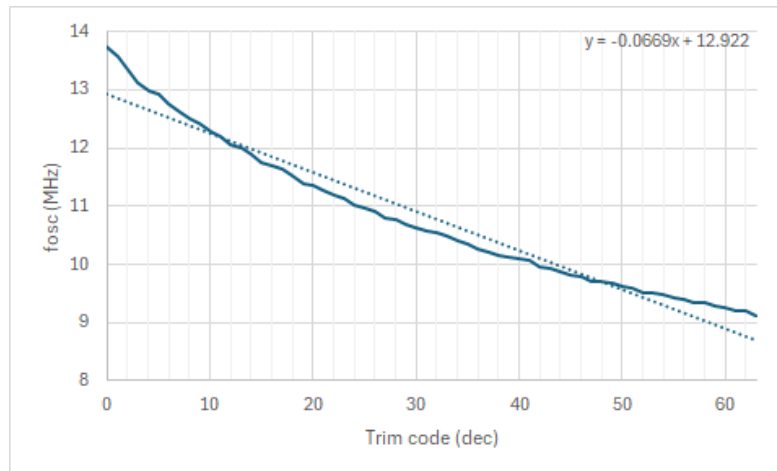
- Results are fully documented at [5] but below shows a summary

- Right shows the oscillator output

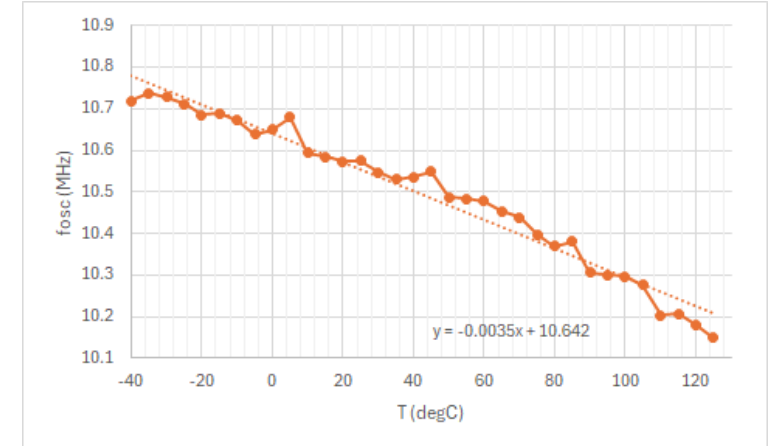


- Right shows the frequency tuning ($\pm 20\%$)

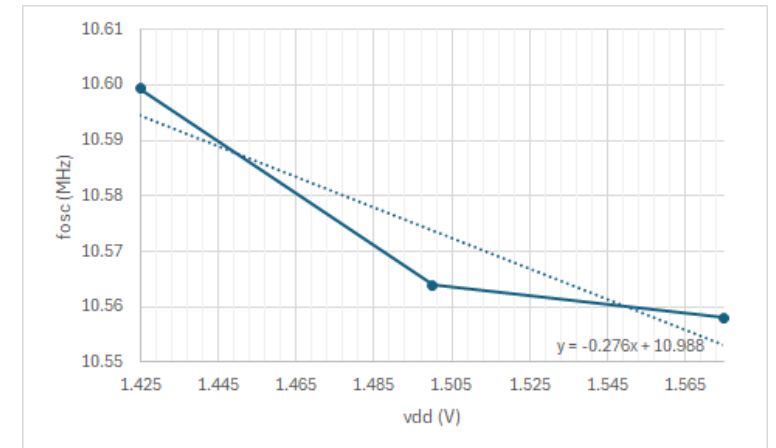
- Note the non-linearity of the CDAC trim ($f_{osc} \propto 1/C_{trim}$)



- Right shows the oscillator temperature sensitivity ($\pm 2.73^\circ\text{C}$ or $3.4\text{kHz}/^\circ\text{C}$)

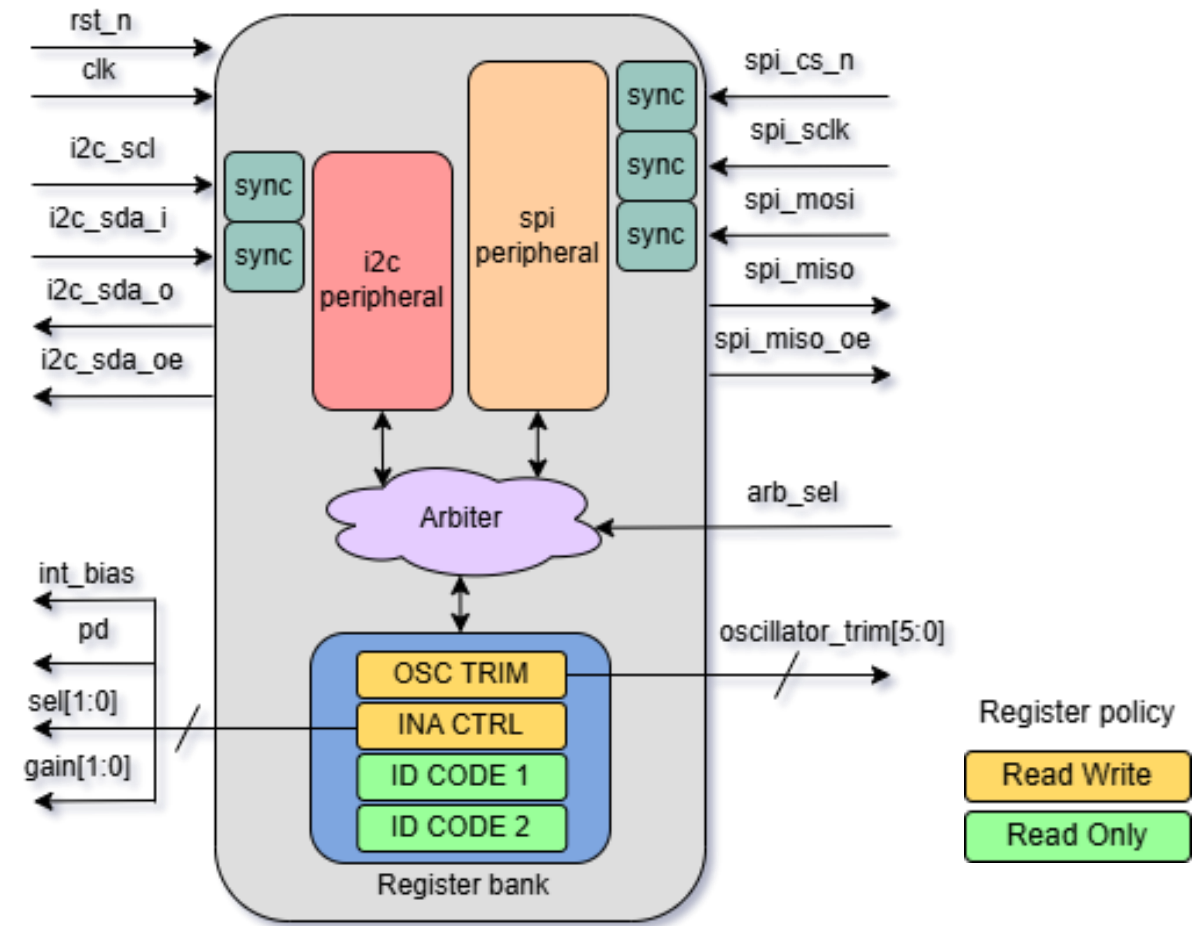


- Right shows the oscillator supply sensitivity ($276\text{Hz}/\text{mV}$)



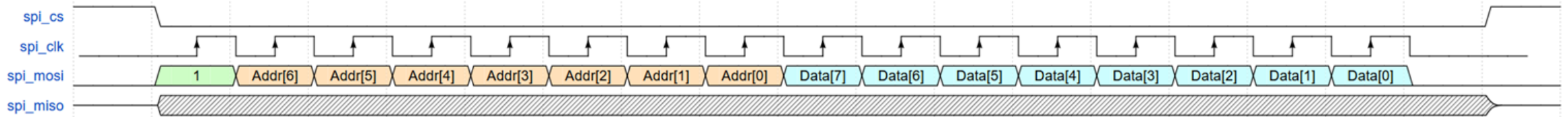
Digital - Architecture

- Synchronous design (oversampling clk)
- CDC sync modules on digital inputs
- SPI Peripheral
 - CPOL/CPHA
 - Frame: 16 bits
 - MISO Tri-state control
- I2C Peripheral
 - Flex address: 3 bits device address
- Register bank
 - Oscillator trim
 - Instrumentation Amplifier control
 - ID Codes

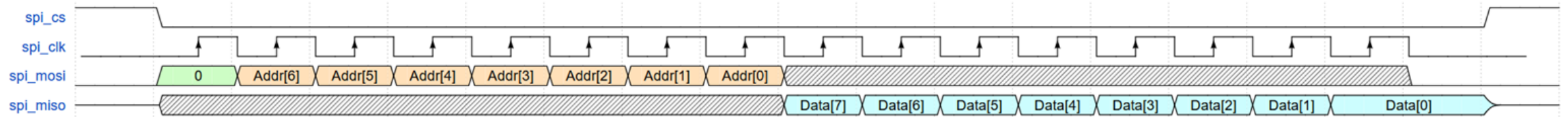


Digital - SPI/I2C Protocol

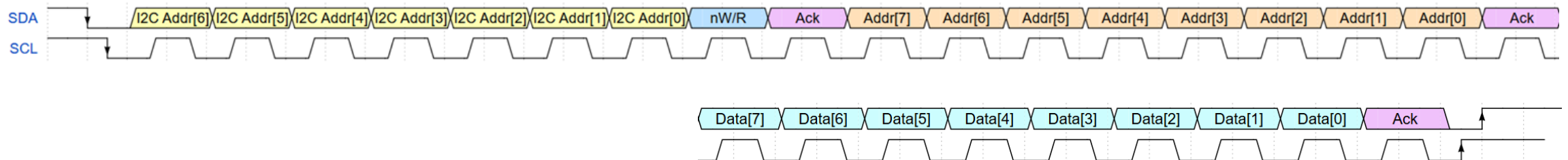
SPI Frame - Write transfer - CPOL=0/CPHA=0



SPI Frame - Read transfer - CPOL=0/CPHA=0

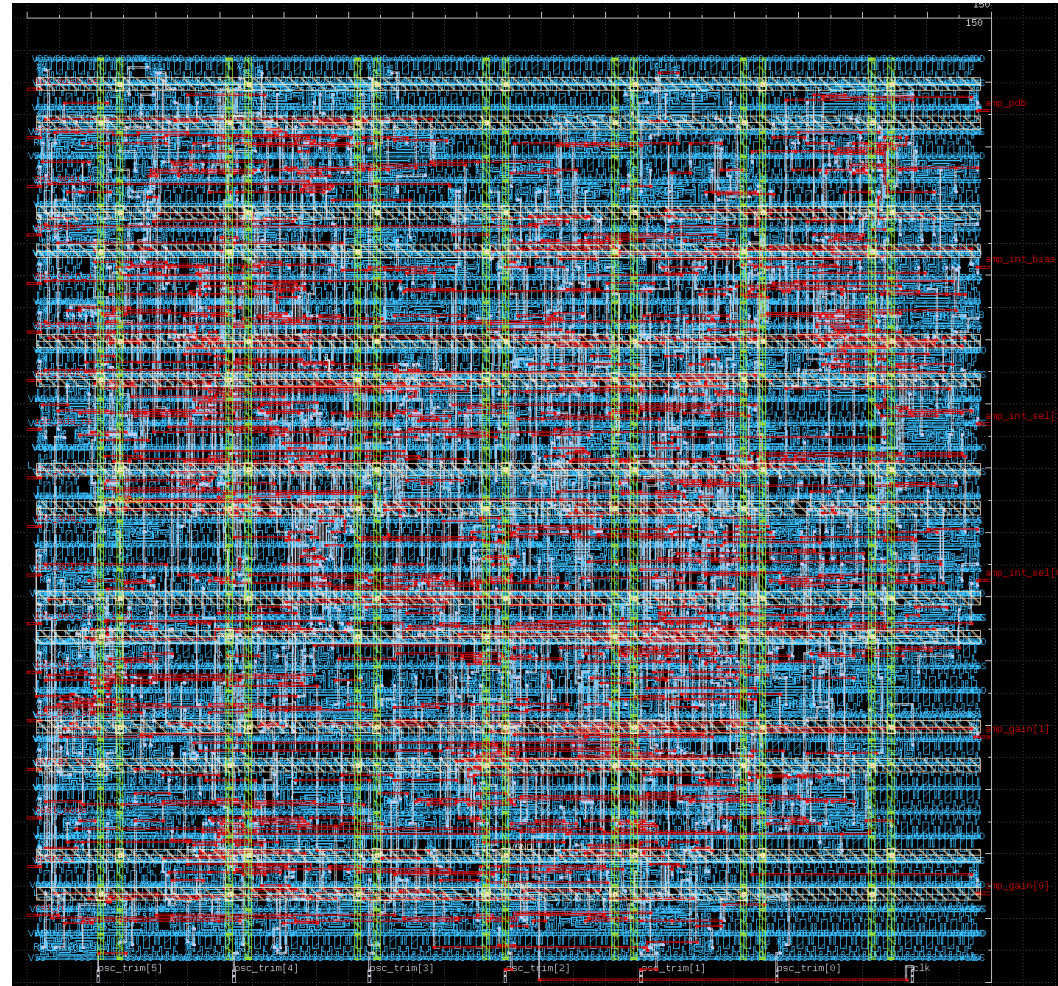


I2C Frame



Digital - Simulation & Physical Implementation

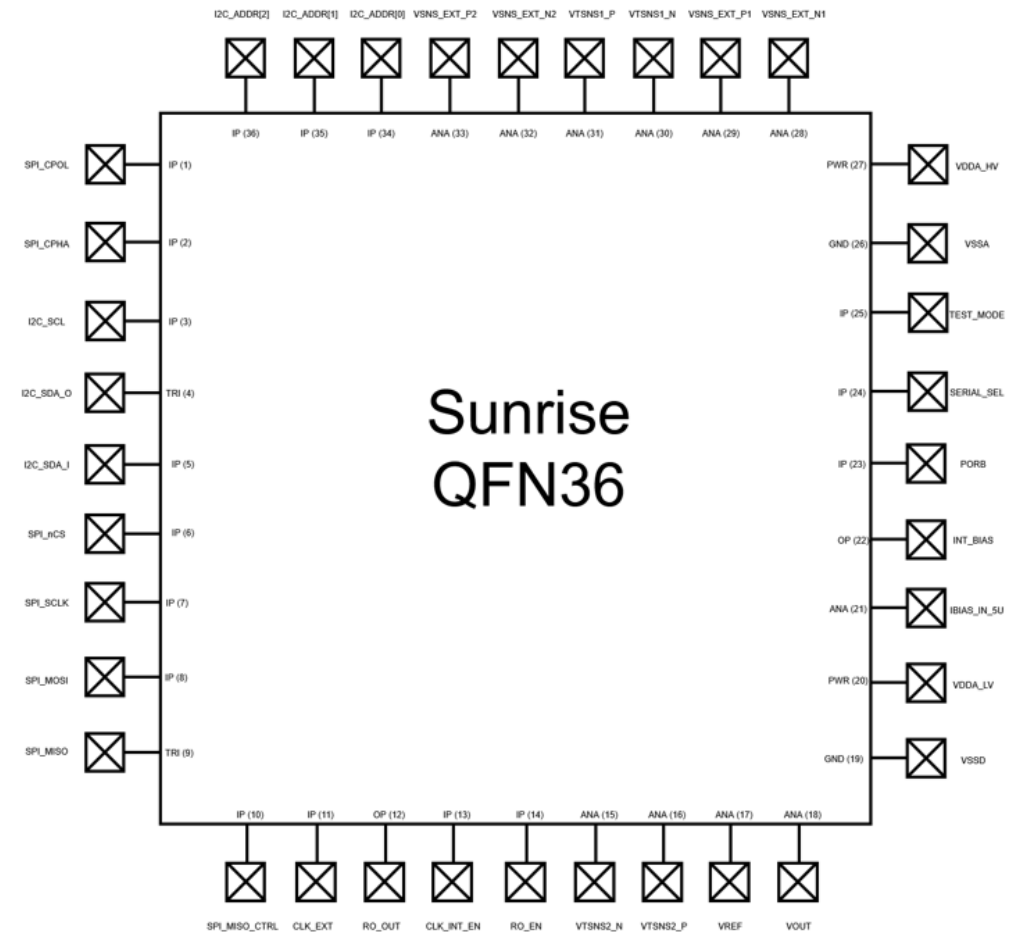
- Digital testbench with cocotb:
 - RTL
 - GL (Verilog netlist / no-timing)
 - Surfer (waveform)
- Physical implementation:
 - Librelane (3.0.3)
 - Target Frequency: 50MHz
 - Max route: M3
 - Die size: 150um²
 - Defined pin cfg
 - Defined PD cfg



Top level pin list

- Shown is the ASIC padding / pinlist definition
- As can be seen, ASIC will be delivered in a QFN36 package
- Note that each pin placement was dictated by the position of its related sub-block in the layout, to avoid antenna violations

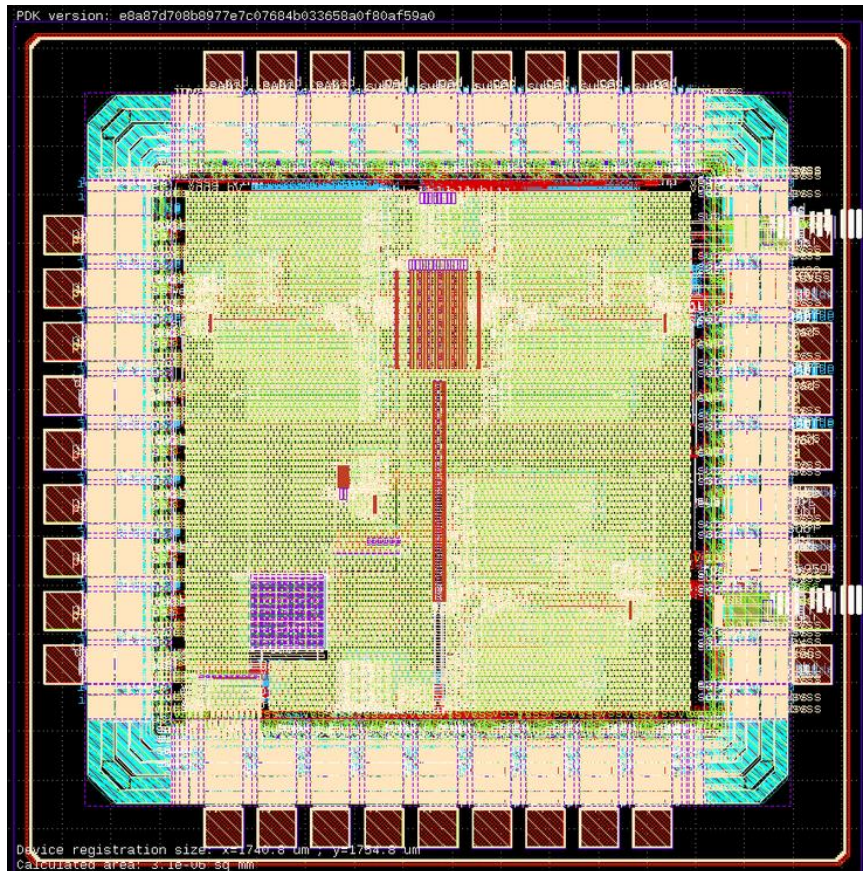
#	Name	Description	IO Type	Domain	Value
1	SPI_CPOL	Mode select (or <i>rdac_ctrl[1]</i>)	IP	Digital	0 / vdda_hv
2	SPI_CPHA	Mode select (or <i>rdac_ctrl[0]</i>)	IP	Digital	0 / vdda_hv
3	I2C_SCL	I2C serial clock lin (or <i>cdac_sel[4]</i>)	IP	Digital	0 / vdda_hv
4	I2C_SDA_O	I2C data out	TRI	Digital	0 / vdda_hv
5	I2C_SDA_I	I2C data in (or <i>cdac_sel[3]</i>)	IP	Digital	0 / vdda_hv
6	SPI_nCS	SPI chip select (or <i>INA_pd</i>)	IP	Digital	0 / vdda_hv
7	SPI_SCLK	SPI serial clock line (or <i>int_bias</i>)	IP	Digital	0 / vdda_hv
8	SPI_MOSI	SPI MOSI data line (or <i>ip_ctrl[1]</i>)	IP	Digital	0 / vdda_hv
9	SPI_MISO	SPI MISO data line	TRI	Digital	0 / vdda_hv
10	SPI_MISO_CTRL	SPI MISO tri-stat ctrl bypass (or <i>ip_ctrl[0]</i>)	IP	Digital	0 / vdda_hv
11	CLK_EXT	External clock	IP	Digital	0 / vdda_hv
12	RO_OUT	RO output	OP	Digital	0 / vdda_hv
13	CLK_INT_EN	Pass RO o/p to digital	IP	Digital	0 / vdda_hv
14	RO_EN	Enable the RO	IP	Digital	0 / vdda_hv
15	VTSNS2_N	Temp sensor bias	ANA	Analog	1 / 10uA
16	VTSNS2_P	Temp sensor bias	ANA	Analog	1 / 10uA
17	VREF	Voltage VOUT is referred to	ANA	Analog	0 – Vref_max
18	VOUT	INA output	ANA	Analog	50mV – 3.3-50mV
19	VSSD	Digital ground	Supply	Digital	0V
20	VDDA_LV	Low voltage supply	PWR	Digital	1.2 – 1.5V
21	IBIAS_IN_5U	External bias	ANA	Analog	5uA
22	INT_BIAS	Sense int_bias internal signal	OP	Digital	0 / vdda_hv
23	PORB	Power on reset (active lo)	IP	Digital	0 / vdda_hv
24	SERIAL_SEL	Select between SPI and I2C (or <i>cdac_sel[5]</i>)	IP	Digital	0 / vdda_hv
25	TEST_MODE	Repurpose SPI / I2C controls as direct i/p's	IP	Digital	0 / vdda_hv
26	VSSA	Analog ground	GND	Analog	0V
27	VDDA_HV	High voltage supply	PWR	Analog	3.3V
28	VSNS_EXT_N1	External sense (-ve)	ANA	Analog	0.9 – Vcm_max_n
29	VSNS_EXT_P1	External sense (-ve)	ANA	Analog	0.9 – Vcm_max_n
30	VTSNS1_N	Temp sensor bias	ANA	Analog	1 / 10uA
31	VTSNS1_P	Temp sensor bias	ANA	Analog	1 / 10uA
32	VSNS_EXT_N2	External sense (-ve) with 260Ω CDM protection	ANA	Analog	0.9 – Vcm_max_n
33	VSNS_EXT_P2	External sense (+ve) with 260Ω CDM protection	ANA	Analog	0.9 – Vcm_max_p
34	I2C_ADDR[0]	Address (or <i>cdac_sel[0]</i>)	IP	Digital	0 / vdda_hv
35	I2C_ADDR[1]	Address (or <i>cdac_sel[1]</i>)	IP	Digital	0 / vdda_hv
36	I2C_ADDR[2]	Address (or <i>cdac_sel[2]</i>)	IP	Digital	0 / vdda_hv



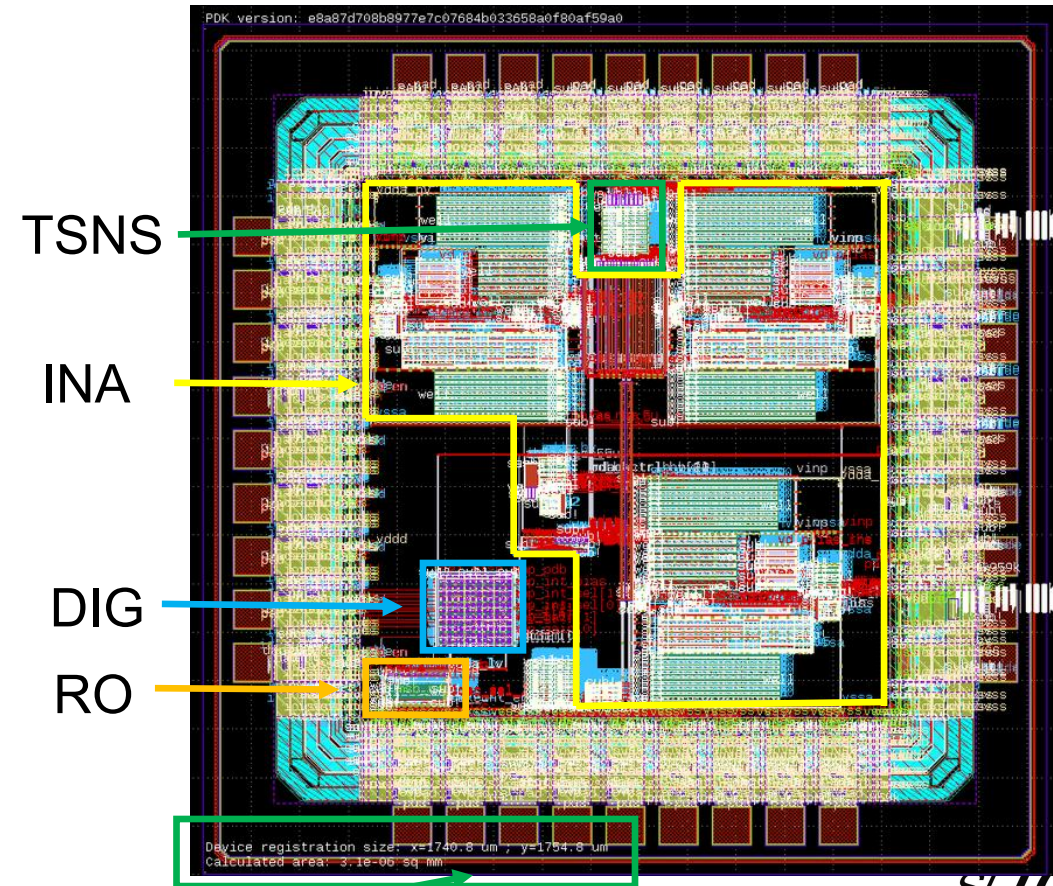
Top level layout

- Shown is the top level ASIC view in klayout

Top level view including the power grid (M4/TM1)



Top level view excluding the power grid (M4/TM1)



What could be improved

- Flow to LVS the digital and analog blocks together
 - There is a way but we were unable to get it working
- Create IO cell schematics which are LVS clean with their corresponding layouts for IHP
 - IO cells can be extracted (done for this project), however no corresponding schematics exist for each IO cell type
 - This would enable a full LVS verification of a completed padding
 - Would enable complete mixed signal sim of the entire ASIC (padding = SCH, analog = RCX/SCH, digital = Verilog)
 - Nice little project for a university student – lots of learnings on ESD / layout / LVS / PEX / OS tools
- Enhance pycell capability
 - Pycells exist for individual devices
 - Why not extend this to matched pairs (diff pairs, cascodes, mirrors)
 - Enable pycells to generate multiple rows of devices and include tap in between each row (currently not supported)
 - In general klayout pycells are very basic. Magic pycells are far superior.
- Create noise summary script for IHP
 - Exists for SKY130 but not seen to be trivial to create for IHP (helps noise debug massively)
- Klayout DRC checker for IHP
 - Seems to be quite inefficient as it takes long time for larger blocks (crashes out for top level runs)
 - Magic faster by orders of magnitude and never crashes out

Conclusions

- Project motivated by the question - is the OS infrastructure ready to produce industry quality products?
- We have demonstrated the development of a low frequency, analog on top ASIC (programmable INA) with industry quality performance, verified to industry standards
- ASIC tapes out on July 13th, returning from the fab on Dec. 16th
- Therefore, we won't be finally able to answer this question (w.r.t low frequency, analog on top products) until post silicon char (~ Feb. '27)
- However for now, based on simulation results, things look positive
- Of course, scaling up to meet customer demands is another requirement
- Low volume products (< 100k parts per year) are only attractive to the OS flow (NRE development costs become less significant for higher volume products)
- Discussions with various suppliers (especially SKY130) have shown that such volumes can be easily met
- Although the OS infrastructure is can't go below 130nm, approx. 2/3's of analog on top chips can be fabricated at these higher, lower costs nodes
- Therefore, it is only a matter of time before the OS infrastructure develops an analog on top, low frequency ASIC for industry with volumes < 100k/yr on either the 130/180nm nodes (if not already done)

Acknowledgements

- It took help from a lot of people to get this project over the line!
- Special thanks to the following:
 - Nlnet for funding the project
 - Alonso for doing such a great job implementing / verifying the digital
 - Iulian Oancea for providing layout guidance and developing the char board
 - Herman Krzysztof for performing regular pre-fab checks on blocks throughout the layout process
 - Roel Jordans for allowing us to use his level shifter IP
 - Stefan Schippers for creating xschem and for updating it according to the demands of this project
 - Holger Vogt / Dietmar Warning for developing ngspice and for addressing various queries I had throughout this project
 - Martin Köhler for creating the align plugin for klayout (and also the move quickly / pad plugins)
 - Yacine Sotehi for fixing the IHP density checker script and implementing the power via insertion script
 - Harald Pretl for introducing us to SERMA
 - Tim Edwards for managing magic, updating it according to the demands of this project and for doing such a great job on the IHP pycells for magic
 - In addition, thanks to Tim for answering my questions which were approximately 3 per day since the beginning of this project! I'm sure I drove him nuts but he always gave an impactful answer to solve my issues
- And finally ...
- Thanks to you – the OS community for your passion, energy and enthusiasm to a highly impactful, potentially game changing movement! Keep it up ... we're all in this together!

References

- [1] <https://www.analog.com/media/en/technical-documentation/data-sheets/ad8223.pdf>
- [2] D. C. Fungueiriño, L. A. Enthoven, J. v. Staveren, M. Babaie and F. Sebastiano, "A Cryo-CMOS Smart Temperature Sensor for the Ultrawide Temperature Range From 5 K to 296 K," in *IEEE Solid-State Circuits Letters*, vol. 9, pp. 29-32, 2026
- [3] https://github.com/SLICESemiconductor/Space_grade_Instrumentation_Amplifier_ASIC/tree/main/Design_Review_1
- [4] https://github.com/SLICESemiconductor/Space_grade_Instrumentation_Amplifier_ASIC/tree/main/Design_Review_2/INA
- [5] https://github.com/SLICESemiconductor/Space_grade_Instrumentation_Amplifier_ASIC/tree/main/Design_Review_2/RO

Thank You

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